

Part III delves into the main tenets of OOP:

- Chapter 8 discusses encapsulation, which involves hiding implementation details from external code and placing all related methods in one place: a class.
- Chapter 9 introduces polymorphism—the idea that multiple classes can have methods with the same names—and shows how it enables you to make calls to methods in multiple objects, without having to know the type of each object. We'll build a Shapes program to demonstrate this concept.
- Chapter 10 covers inheritance, which allows you to create a set of subclasses that all use common code built into a base class, rather than having to reinvent the wheel with similar classes. We'll look at a few real-world examples where inheritance comes in handy, such as implementing an input field that only accepts numbers, then rewrite our Shapes example program to use this feature.
- Chapter 11 wraps up this part of the book by discussing some additional important OOP topics, mostly related to memory management. We'll look at the lifetime of an object, and as an example we'll build a small balloon-popping game.

Part IV explores several topics related to using OOP in game development:

- Chapter 12 demonstrates how we can rebuild the card game developed in Chapter 1 as a pygame-based GUI program. I also show you how to build reusable `Deck` and `Card` classes that you can use in other card games you create.
- Chapter 13 covers timing. We'll develop different timer classes that allow a program to keep running while concurrently checking for a given time limit.
- Chapter 14 explains animation classes you can use to show sequences of images. We'll look at two animation techniques: building animations from a collection of separate image files and extracting and using multiple images from a single sprite sheet file.